SDS188 A1S2 — FINAL CRAM MARKS-RESCUE MAP

COLOUR KEY: GREEN = do/use · RED = trap/never · ORANGE = decision/tails · PURPLE = memory · BLUE = Excel/formula

1) FIRST 90 SECONDS — WRITE THIS ON ROUGH PAPER

TAIL: H1 > → RIGHT → 1−area · H1 < → LEFT · H1 ≠ → TWO tails
Z/t: σ known → Z. σ unknown + s → t. df=n−1.
CI: estimate ± critical×SE. 95% → α=.05 → .025 each tail.
ERRORS: Type I = reject true H0 = α. Type II = miss false H0 = β.
Power=1−β.
MEMORY: H1 gives TAIL → σ gives Z/t → α gives CUT → compare → decision.

2) EXCEL FUNCTION DECODER — NEVER MIX THESE

DIST = value → probability. INV = probability → boundary.
NORM.DIST(x;μ;SD;TRUE) = area LEFT of x.
NORM.INV(p;μ;SD) = boundary with p LEFT.
NORM.S.INV(p) = z critical.
T.DIST(t;df;TRUE) = area LEFT of t.
T.INV(p;df) = t boundary, left area p.
T.INV.2T(α;df) = positive TWO-tail critical t.
TRAP: Excel on your Mac uses decimal comma + semicolon; EMSLearn answers use full stop + 4 decimals.

3) BOOTSTRAP — PRACTICE Q1

When? Small sample + no idea of population distribution.
Story: original sample → resample WITH replacement, same n → mean each resample → repeat → order MEANS → cut tails.
Excel: each resample row: =AVERAGE(D2:W2) → fill down 250 rows.
95% CI → 2.5% each tail. Limits come from ordered bootstrap means, NOT original data.
Never use min/max; never automatically use z/t CI; do not resample without replacement.
MEMORY: ROWS → MEANS → TAILS.

4) PROPORTION CI — PRACTICE Q2

Sample proportion: p̂=x/n = successes ÷ total.
SE: √[p̂(1−p̂)/n]. Excel: =SQRT(p*(1-p)/n)
CI: p̂ ± z*SE. 95% z*=NORM.S.INV(.975)≈1.96.
Upper: p̂ + z*SE · Lower: p̂ − z*SE.
TRAP: if question says “ASSUME p̂=... and SE=...” for later part, SWITCH to those supplied values.
MEMORY: p-hat in the MIDDLE; z×SE on BOTH sides.

5) t CRITICAL + DECISION — PRACTICE Q3

H1: μ>μ0 → RIGHT tail. s given / σ unknown → t. df=n−1.
Right critical: =T.INV(1-α;df)
Left critical: =T.INV(α;df)
Two-tail: =T.INV.2T(α;df)
Right-tail decision: tcalc > tcrit → REJECT H0; otherwise do not reject.
Never say “accept H0”. MEMORY: s → t; n−1; arrow tells tail.

6) Z TEST + CRITICAL VALUE IN SECONDS + POWER — Q4

σ GIVEN → Z. H1: μ>μ0 → right tail.
SE: σ/√n. Critical z: NORM.S.INV(1−α).
Critical value IN ORIGINAL UNITS:
x̄crit = μ0 + zcrit(σ/√n)
or directly: =NORM.INV(1-α;μ0;σ/SQRT(n))
TRAP: if question says “in seconds”, zcrit alone is NOT the answer.

POWER at true μ1: rejection region is x̄ > x̄crit.
=1-NORM.DIST(xbar_crit;μ1;σ/SQRT(n);TRUE)
β = NORM.DIST(xbar_crit;μ1;SE;TRUE); Power=1−β.
MEMORY DIAGRAM: H0 centre → |critical line| → reject →. Move centre to true μ1; area beyond line = POWER.

7) SAMPLE SIZE — PROPORTION + FINITE POPULATION

Proportion desired margin E:
n = z² p(1−p)/E². If p unknown use p=.5 (most conservative).
=ROUNDUP((NORM.S.INV(1-α/2)^2*p*(1-p))/E^2;0)
ALWAYS round sample size UP.

Finite population correction: if population N is finite and course asks adjusted n, use the course finite-population formula from your sheet; identify N, confidence, E and variability first.
TRAP: sample-size answers are whole people/items → ROUND UP, never ordinary rounding down.

8) TYPE I / TYPE II / POWER — FAST CONCEPT MARKS

Type I: H0 true but reject it → probability α.
Type II: H0 false but fail to reject → probability β.
Power: correctly reject false H0 = 1−β.
If confidence level ↓ (e.g. 99%→95%), α ↑ → rejection easier → β ↓ → power ↑.
MEMORY: I = false alarm; II = missed alarm.

9) UNKNOWN σ: PROBABILITY FOR SAMPLE MEAN — Q8 STYLE

Population normal + σ unknown + sample s given → t distribution.
SE=s/√n; df=n−1.
For P(L<x̄<U): calculate tL=(L−μ)/(s/√n), tU=(U−μ)/(s/√n).
Excel: =T.DIST(tU;df;TRUE)-T.DIST(tL;df;TRUE)
MEMORY: BETWEEN = upper LEFT-area − lower LEFT-area.

10) CH6/7 SAFETY NET — EASY RESCUE MARKS

Sample mean distribution: mean=μ; SE=σ/√n.
P(x̄<x)=NORM.DIST(x;μ;σ/SQRT(n);TRUE)
P(x̄>x)=1−that.
Boundary from probability: NORM.INV(left_area;μ;σ/SQRT(n))
Central C interval around μ: tails=(1−C)/2; lower NORM.INV(tail;μ;SE), upper NORM.INV(1−tail;μ;SE).
TRAP: sample MEAN problems use σ/√n, not σ.

11) SANITY CHECKS + MARK-RESCUE HACKS

① Units check: z/t has no units; “seconds/apps/grams” boundary must be in original units.
② Probability check: probability/power/β must be between 0 and 1.
③ CI check: lower < estimate < upper.
④ SE check: larger n → smaller SE.
⑤ Tail check: draw arrow from H1 before touching Excel.
⑥ Function check: need probability? DIST. Need boundary? INV.
⑦ Raw data: immediately make helper cells: n=COUNT(range), xbar=AVERAGE(range), s=STDEV.S(range). Only use s if σ is NOT supplied.
⑧ Keep full precision in Excel; round only final EMSLearn entry.
⑨ If stuck, write GIVEN → FIND → TAIL → Z/t → FORMULA. This often unlocks the question.

FINAL SAY-ALOUD SCRIPT: “H1 gives the tail. Sigma tells me Z or t. DIST gives probability; INV gives boundary. CI is middle ± critical×SE. Power is 1−beta. Check units, check tails, then enter 4 decimals.”